UTS	SCHOOL OF COMPUTING AND CREATIVE MEDIA BACHELOR OF COMPUTER SCIENCE (HONS.)
	Title Registration Form (Final Year Project)

Details of Student				
Student Name:	VINCENT LEE CHENG HONG	Project Category (tick one):	■	System/Application Development
Student ID:	BCS24020018		☐	Research-Based
Phone Num:	011-1400 4966		☐	Hybrid (Research + Development)
Programme:	Bachelor of Computer Science (Hons)	Proposed Project Title:	AI Local News Summarization and Alert Bot on Telegram	
Semester / Academic Year:	FEB 2025/2026			

Details of Supervisor		**Details of Co-Supervisor (optional)**	
Name:	Ms. Khairunnisa Ibrahim	Name:	
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Signature of Student:

Name: VINCENT LEE CHENG HONG

Date: 24/02/2026

Acknowledged by:

Name of Supervisor: Ms. KHAIRUNNISA IBRAHIM

Date: 27/2/2026

<table>
<tr><th>Project Title & Short Description</th><th>Current Problem/Pain Point/Challenges</th><th>Value Brought by the Project</th></tr>
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<td>Intelligent Local News Summarization and Alerting Bot on Telegram Using Local LLM.
A backend-driven automated system that scrapes local news, processes it using a locally hosted AI to generate short summaries, and pushes scheduled updates directly to users via a Telegram Bot.
It aims to bridge the information gap for younger generations and busy professionals by delivering relevant community news without information overload.</td>
<td>Fragmented Sources: Local news is often trapped in closed social media platform like Facebook, making it inaccessible to users without active accounts.
Information Overload: Mainstream apps prioritize global headlines, lack of critical community updates like utility cuts or local news.
Inefficient Consumption: Local reports are often lengthy and unstructured, preventing busy professionals from quickly extracting key information.</td>
<td>Enhanced Community Awareness (SDG 11): Increases local engagement by ensuring residents are informed about safety, infrastructure, events, and news.
Time Efficiency: Saves users time by providing AI-generated summaries in Telegram chat.
Privacy & Cost-Effectiveness: Utilizes a Local LLM to process data, eliminating subscription costs for third-party AI APIs.</td>
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<tr><th>Scope (Target user, tools, etc.)</th><th>Similar System/Application/Research</th><th>Proposed Project Features</th></tr>
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<td>Target Users: Young adults, busy working professionals, and local community residents in Sarawak.
Platform: Telegram Messaging App.
Tools/Technologies:
Tools: Telegram BotFather API.
Backend: Python.
AI: Ollama (Llama 3.1) for summarization.
Data: RSS Feeds, Web Scrapers.
Limitations: The bot requires the host machine (my laptop) to be online to process and push messages; it does not feature a custom graphical user interface (GUI).</td>
<td>Google News: Excellent for global/national news but lacks "local" sources like specific area updates.
Facebook Community: Rich in local info but unstructured, filled with noise or spam, and requires an active account to browse.
Official WhatsApp Channels: Delivers updates directly, but requires users to subscribe to multiple publishers individually.
Standard Telegram RSS Bots: User pastes an RSS link, and the bot automatically forwards every new article to the chat</td>
<td>Automated Aggregation: Scrapes and compiles news from defined local sources such as RSS feeds, local websites.
AI Summarization: Automatically converts long articles into 30-word briefs or bullet points using a Local LLM.
User Personalization: Allows users to configure their bot settings to filter news by specific categories, locations, and preferred push frequency.
Urgency Notification: Analyzes the news to send push notifications only for high-priority local incidents such as floods.</td>
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